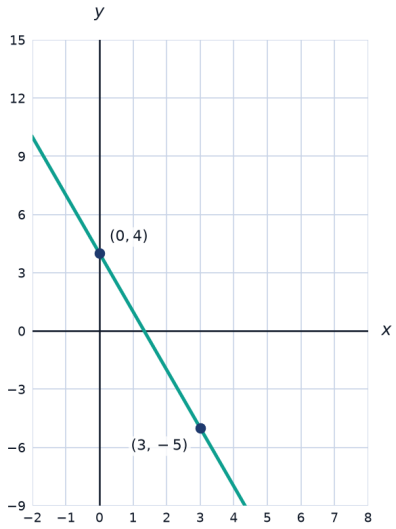
# Linear Functions & Inequalities

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

## Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

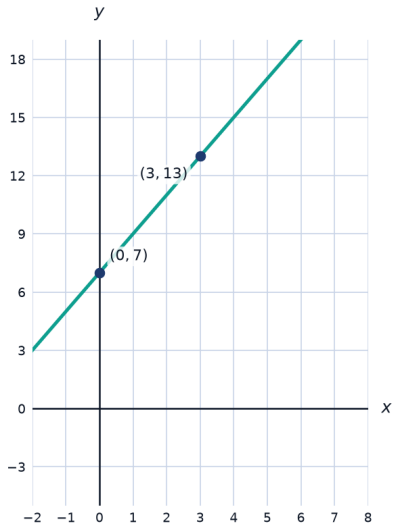
**1** [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the municipal library, is shown in the $xy$-plane. Based on the two labeled points on the line, what is the value of $f(6)$?




**2** [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

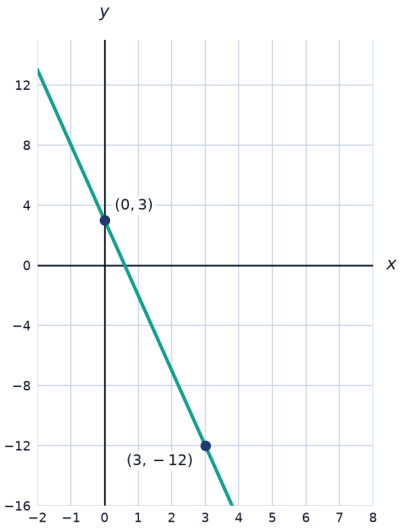
The graph of the linear function $y = f(x)$, which models the daily balance index at the coastal aquarium, is shown in the $xy$-plane. Based on the two labeled points on the line, what is the value of $f(6)$?




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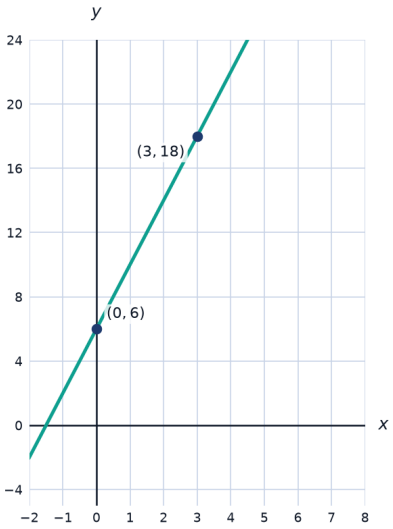
**3** [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the urban bike-share station, is shown in the $xy$-plane. Based on the two labeled points on the line, what is the value of $f(6)$?




**4** [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

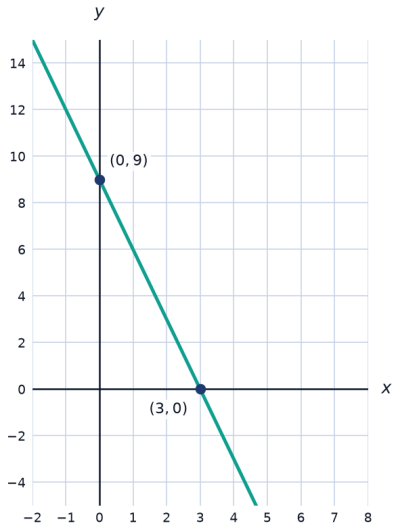
The graph of the linear function $y = f(x)$, which models the daily balance index at the agricultural cooperative, is shown in the $xy$-plane. Based on the two labeled points on the line, what is the value of $f(6)$?




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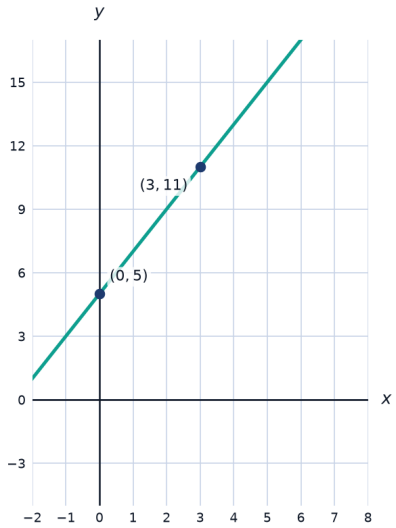
**5** [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the recycling center, is shown in the $xy$-plane. Based on the two labeled points on the line, what is the value of $f(6)$?




**6** [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

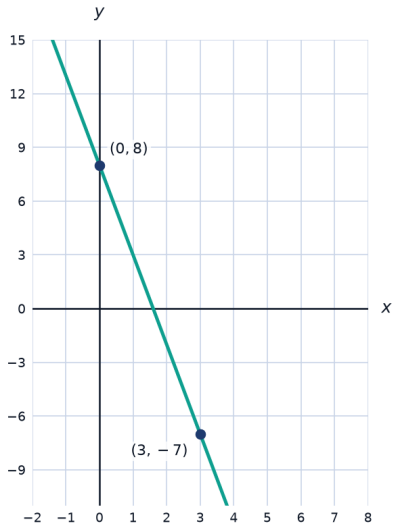
The graph of the linear function $y = f(x)$, which models the daily balance index at the neighborhood bookstore, is shown in the $xy$-plane. Based on the two labeled points on the line, what is the value of $f(6)$?




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**7** [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the culinary institute, is shown in the $xy$-plane. Based on the two labeled points on the line, what is the value of $f(6)$?




## Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

**8** [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the maple syrup farm, what is the slope of the line in the $xy$-plane that passes through the points $(-3, 5)$ and $(1, -3)$?

**9** [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the marine research vessel, in a calculation for the local farmers market (case 20), if $f(x) = 7x + 6$, what is $f(3)$?

**10** [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the campus coffee roastery, in a calculation for the performing arts academy (case 24), if $f(x) = -7x - 4$, what is $f(6)$?

**11** [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the meteorological station, in a calculation for the coastal aquarium (case 28), if $f(x) = 7x - 4$, what is $f(1)$?

**12** [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the urban bike-share station, what is the slope of the line passing through $(2, 5)$ and $(6, 17)$?

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**13** [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the art gallery, in the coordinate plane, the graph of $y = H(y)$ is translated $8$ units right and $9$ units up to form $y = K(y)$, mapping $E(12, 61)$ to $F$. What is the sum of coordinates of $F$ on $y = K(y)$?

**14** [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the sports complex, a line is parallel to the line $y = 2x + 3$ and passes through the point $(7, -4)$. What is its $y$-intercept?

## Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

**15** [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the regional airport shuttle, let the functions $U$ and $V$ be defined by $U(v) = 23v - 15$ and $V(v) = v^2 + 7$. What is the composite value $U(V(7))$?

**16** [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The botanical conservatory orders adult tickets at USD $4$ per unit plus a flat shipping fee of USD $15$. If the total order cost cannot exceed USD $38$, what is the maximum whole number of adult tickets that can be ordered?

**17** [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In the $xy$-plane map of the municipal library, line $k$ is perpendicular to the line $y = 3x + 3$ and passes through the point $(6, 6)$. What is the $y$-intercept of line $k$?

**18** [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A water storage tank at the community theater initially holds $130$ gallons and is filled at a constant rate of $10$ gallons per minute. How many gallons of water are in the tank after $6$ minutes?

**19** [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The regional bakery orders standard passes at USD $7$ per unit plus a flat shipping fee of USD $18$. If the total order cost cannot exceed USD $80$, what is the maximum whole number of standard passes that can be ordered?

**20** [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

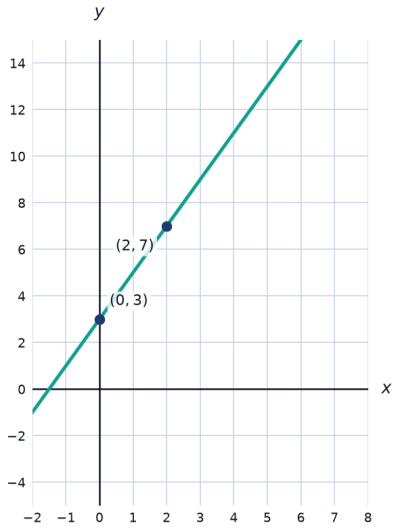
In the $xy$-plane map of the coastal aquarium, line $k$ is perpendicular to the line $y = 6x + 6$ and passes through the point $(12, 9)$. What is the $y$-intercept of line $k$?

## Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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**21** [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the botanical conservatory, is shown in the $xy$-plane. Based on the two labeled points on the line, what is the value of $f(5)$?




**a.** 13

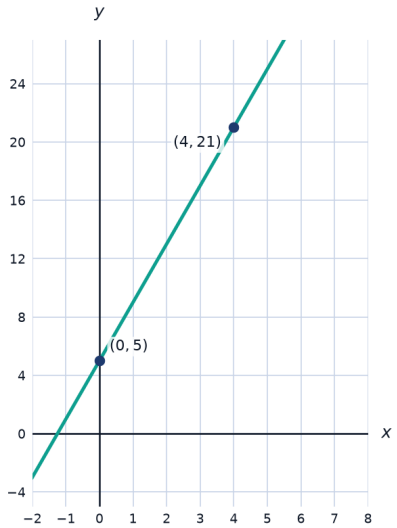
**b.** 15

**c.** 10

**d.** 16

**22** [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the community theater, is shown in the $xy$-plane. Based on the two labeled points on the line, what is the value of $f(7)$?




**a.** 28

**b.** 26

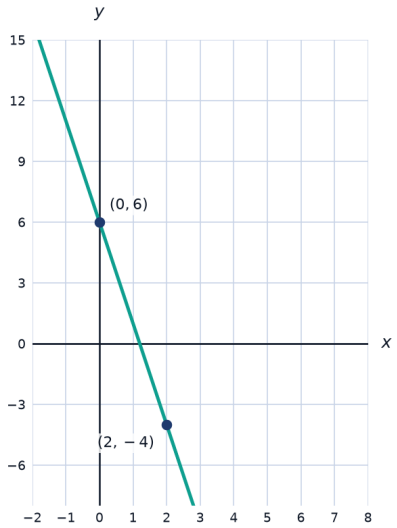
**c.** 33

**d.** 37

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**23** [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the regional bakery, is shown in the $xy$-plane. Based on the two labeled points on the line, what is the value of $f(5)$?




**a.** 2

**b.** −19

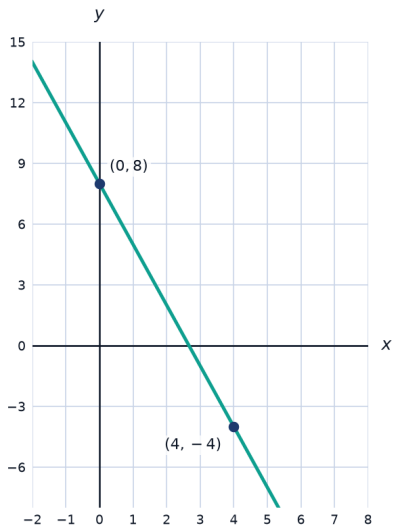
**c.** −24

**d.** −25

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**24** [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the mountain observatory, is shown in the $xy$-plane. Based on the two labeled points on the line, what is the value of $f(7)$?




**a.** $-16$

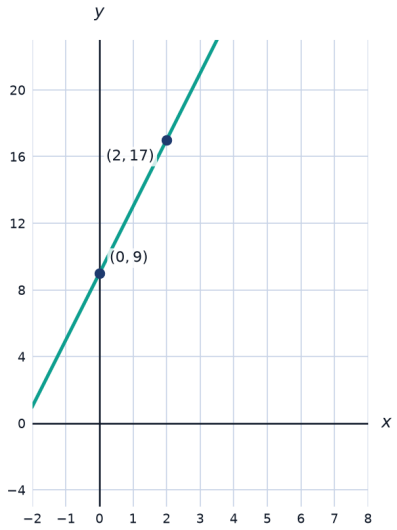
**b.** $-21$

**c.** $4$

**d.** $-13$

**25** [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the science museum, is shown in the $xy$-plane. Based on the two labeled points on the line, what is the value of $f(5)$?




**a.** $20$

**b.** $26$

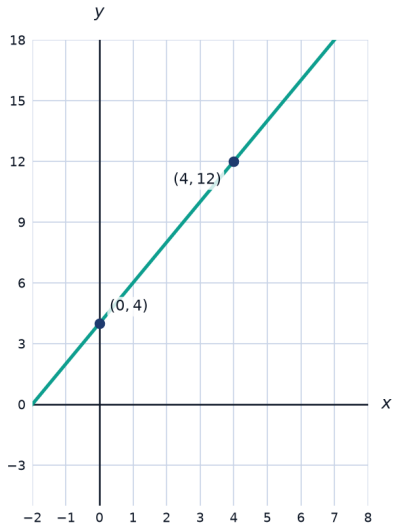
**c.** $29$

**d.** $33$

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**26** [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the solar microgrid facility, is shown in the $xy$-plane. Based on the two labeled points on the line, what is the value of $f(7)$?




**a.** 18

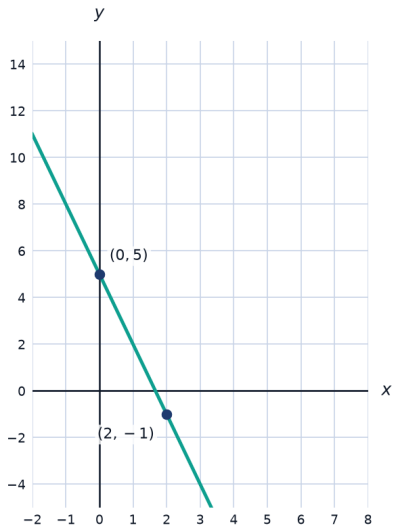
**b.** 20

**c.** 14

**d.** 16

**27** [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the ceramic studio, is shown in the $xy$-plane. Based on the two labeled points on the line, what is the value of $f(5)$?




**a.** −13

**b.** −15

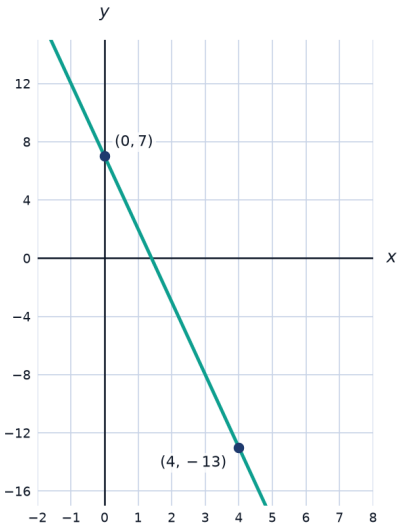
**c.** 4

**d.** −10

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**28** [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the publishing house, is shown in the $xy$-plane. Based on the two labeled points on the line, what is the value of $f(7)$?




**a.** $-6$

**b.** $-28$

**c.** $-33$

**d.** $-35$

## Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

**29** [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the art gallery, in a calculation for the wildlife sanctuary (case 11), if $f(x) = 7x - 11$, what is $f(-9)$?

**a.** $-67$

**b.** $-74$

**c.** $-83$

**d.** $-81$

**30** [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the robotics club, which of the following lines is parallel to the line $y = 4x - 9$ in the $xy$-plane?

**a.** $y = 4x + 12$

**b.** $y = -4x - 9$

**c.** $y = -\frac{1}{4}x + 3$

**d.** $y = \frac{1}{4}x - 9$

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**31** [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the culinary institute, what is the slope of any line in the $xy$-plane that is perpendicular to the line $y = -\frac{2}{5}x + 3$?

**a.** $\frac{5}{2}$

**b.** $-\frac{5}{2}$

**c.** $\frac{2}{5}$

**d.** $-\frac{2}{5}$

**32** [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the water treatment plant, a line has slope $-8$ and passes through the point $(3, -9)$. What is the $y$-intercept of the line?

**a.** $24$

**b.** $9$

**c.** $20$

**d.** $15$

**33** [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the sports complex, in the coordinate plane, the graph of $y = P(r)$ is translated $13$ units right and $3$ units up to form $y = Q(r)$, mapping $U(6, 67)$ to $V$. What is the sum of coordinates of $V$ on $y = Q(r)$?

**a.** $76$

**b.** $89$

**c.** $92$

**d.** $105$

**34** [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the planetarium, a gym charges a monthly membership fee of USD $30$ plus USD $2$ for each fitness class attended. If a member attends $c$ fitness classes in a month, which expression represents the member's total cost for that month?

**a.** $30 + 2c$

**b.** $30c + 2$

**c.** $2(30 + c)$

**d.** $\frac{30 + c}{2}$

**35** [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the regional airport shuttle, the point $(4, 14)$ lies on the line $y = kx - 2$ in the $xy$-plane, where $k$ is a constant. What is the value of $k$?

**a.** $3$

**b.** $4$

**c.** $5$

**d.** $6$

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**36** [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the wind energy cooperative, what is the slope of the horizontal line defined by $y = 7$ in the $xy$-plane?

**a.** 0

**b.** 1

**c.** 7

**d.** undefined

**37** [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the national park visitor center, a line has slope $-7$ and passes through the point $(-2, 12)$. What is the $y$-intercept of the line?

**a.** $-9$

**b.** $-2$

**c.** 1

**d.** $-7$

## Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

**38** [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the maple syrup farm, a line is perpendicular to $y = -\frac{2}{3}x + 5$ and has a $y$-intercept of $(0, -4)$. What is the equation of this line?

**a.** $y = \frac{3}{2}x - 4$

**b.** $y = -\frac{3}{2}x - 4$

**c.** $y = \frac{2}{3}x - 4$

**d.** $y = -\frac{2}{3}x - 4$

**39** [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the planetarium, which point lies in the solution set of the system of inequalities? $\{y > 2x - 3; y \leq -x + 4$

**a.** $(2, 5)$

**b.** $(1, 2)$

**c.** $(3, 2)$

**d.** $(0, -4)$

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**40** [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the wind energy cooperative, line $k$ passes through $(-3, 7)$ and $(5, -9)$. What is the $x$-intercept of line $k$?

**a.** $(0.5, 0)$

**b.** $(1, 0)$

**c.** $(0, 1)$

**d.** $(2, 0)$

**41** [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the marine research vessel, which of the following lines is perpendicular to the line $5x + 2y = 10$?

**a.** $2x - 5y = 15$

**b.** $5x - 2y = 10$

**c.** $2x + 5y = 20$

**d.** $-5x - 2y = 8$

**42** [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The water treatment plant orders fiction novels at USD $4$ per unit plus a flat shipping fee of USD $36$. If the total order cost cannot exceed USD $99$, what is the maximum whole number of fiction novels that can be ordered?

**a.** 16

**b.** 14

**c.** 17

**d.** 15

**43** [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In the $xy$-plane map of the orchestra hall, line $k$ is perpendicular to the line $y = 4x + 3$ and passes through the point $(8, 27)$. What is the $y$-intercept of line $k$?

**a.** 33

**b.** 27

**c.** 29

**d.** 25

**44** [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A water storage tank at the sports complex initially holds $235$ gallons and is filled at a constant rate of $13$ gallons per minute. How many gallons of water are in the tank after $9$ minutes?

**a.** 365

**b.** 352

**c.** 244

**d.** 339

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**45** [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The charter ferry service orders hiking packs at USD $7$ per unit plus a flat shipping fee of USD $39$. If the total order cost cannot exceed USD $94$, what is the maximum whole number of hiking packs that can be ordered?

**a.** 7

**b.** 8

**c.** 6

**d.** 9

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